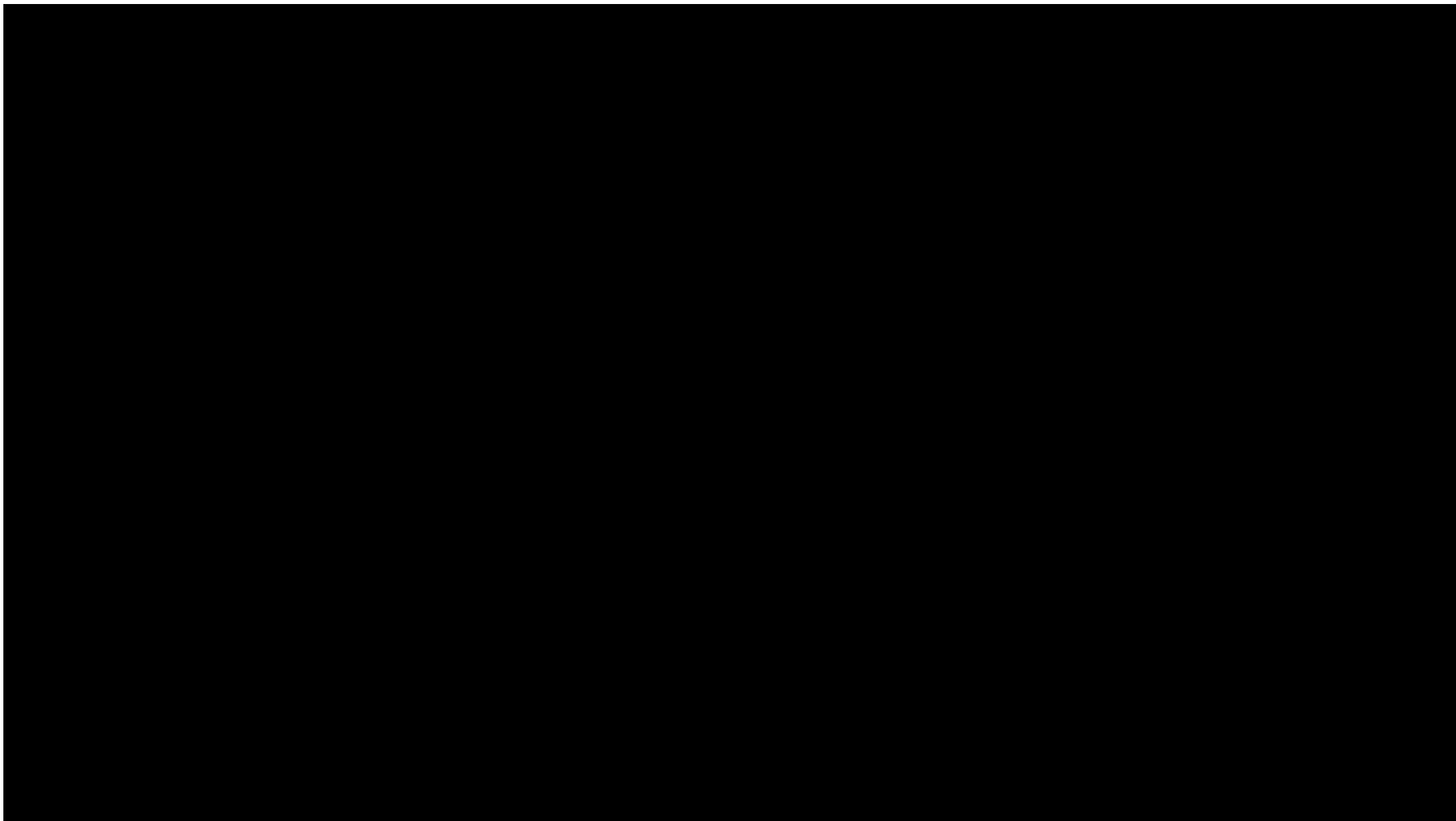




Data

images /
texts /
audio



human-crafted
features



ML

→ prediction

images: HoG, SIFT, texton-boost, ...
texts: TF-IDF, BM25, ...
audio: Spectrograms

Data



Neural Network



prediction

end-to-end training

Logistic Regression

$$\{x_i, y_i\}_{i=1}^N, \quad x_i \in \mathbb{R}^d, \quad y_i \in \{-1, 1\}$$
$$y_i \in \{1, 2, \dots, K\}$$

$$y(x) = \text{sign}(w^T x + b)$$

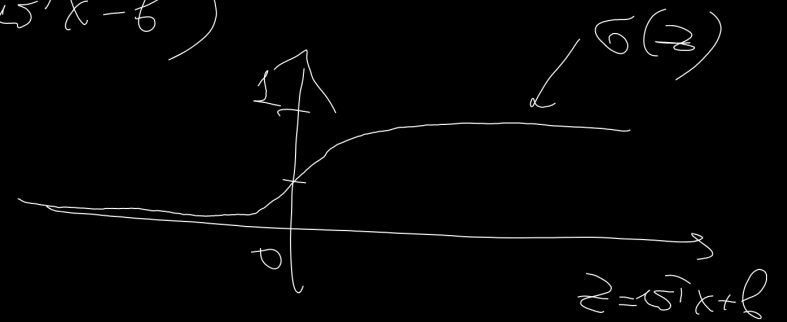
$$y(x) = \underset{k \in \{1, \dots, K\}}{\text{argmax}} \left(\overset{1 \times d}{w_k^T} \overset{d \times 1}{x} + \overset{1 \times 1}{b_k} \right)$$

$$p(y=+1 | w, x, b) = \frac{1}{1 + \exp(-w^T x - b)} = \sigma(w^T x + b)$$

$$p(y=k | x, W, b) =$$

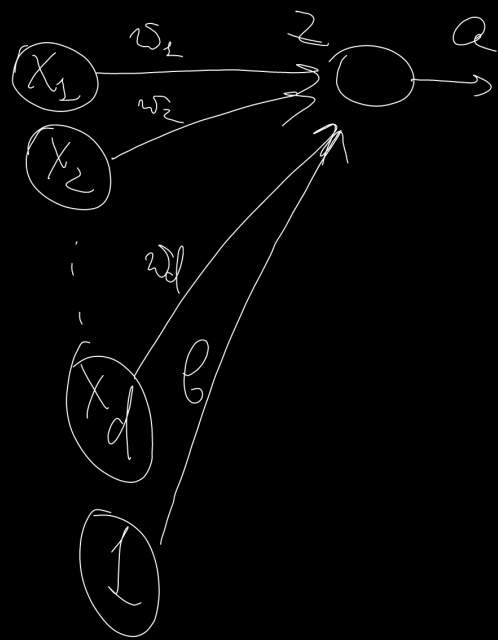
$$= \frac{\exp(w_k^T x + b_k)}{\sum_{j=1}^K \exp(w_j^T x + b_j)}$$

$$= \text{SoftMax}(Wx + b)$$



$$p(y_1, \dots, y_N | x_1, \dots, x_N, w, b) = \prod_{i=1}^N p(y_i | x_i, w, b) \rightarrow \max_{w, b}$$

$$-\log p(y_1, \dots, y_N | x_1, \dots, x_N, w, b) = -\sum_{i=1}^N \log p(y_i | x_i, w, b) \rightarrow \min_{w, b}$$

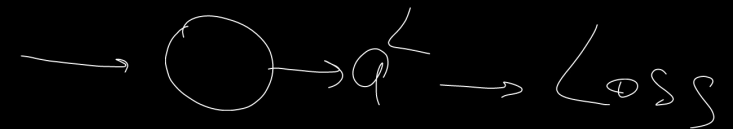
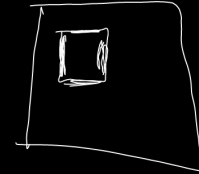
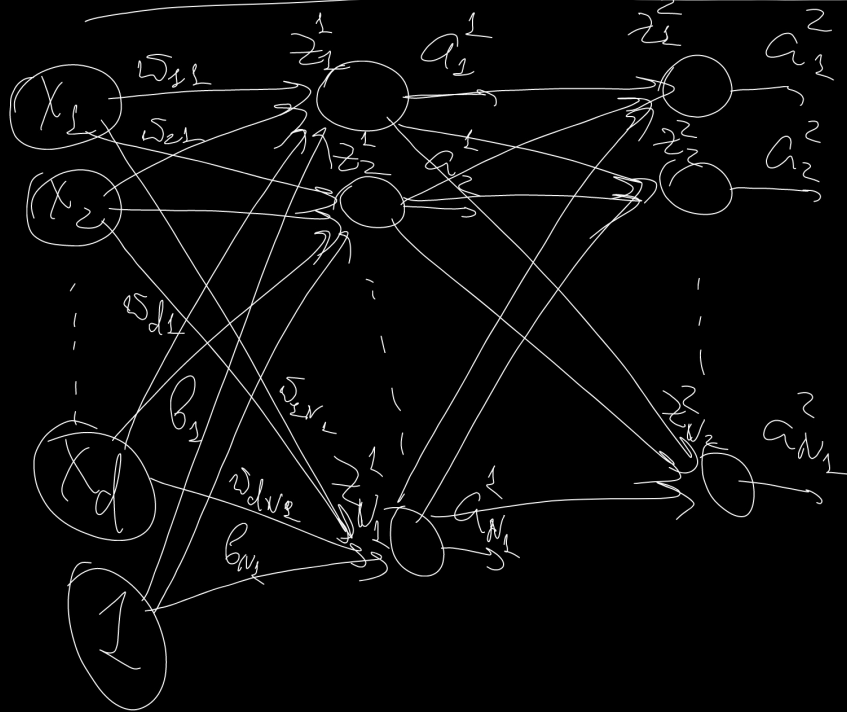


$$z = w^T x + b$$

$$a = \sigma(z)$$

↑
diff. and convex

Multi Layer Perceptron (MLP)



MLP forward pass

$$a^0 := x$$

for $l = 1, 2, \dots, L$:

$$z^l = W^l a^{l-1} + b^l$$

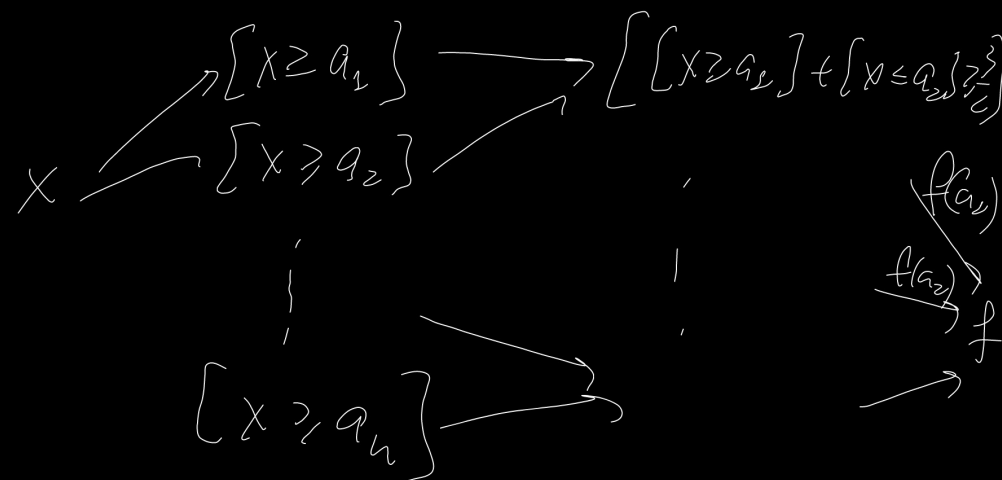
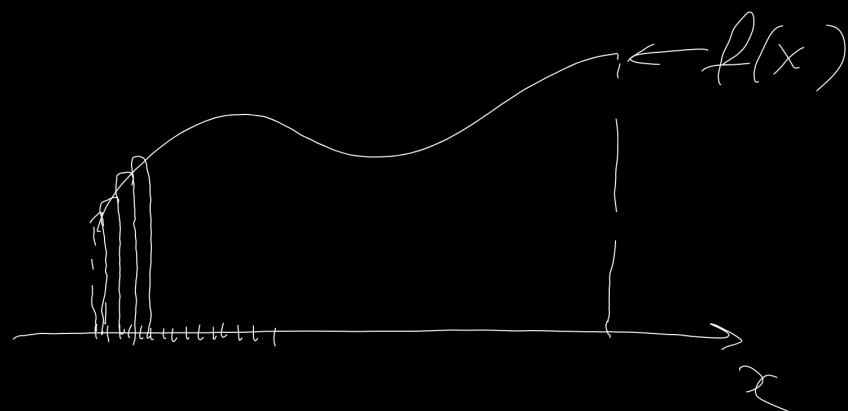
$$a^l = \sigma(z^l)$$

output $\text{Loss}(a^L)$

$$\sum_{i=1}^N \text{Loss}(y_i, a^L(x_i, \theta))$$

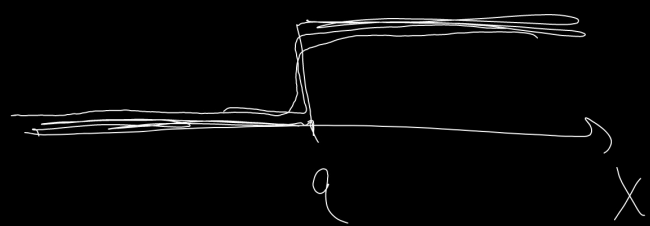
$\rightarrow \text{min}_{\theta}$

$$\theta = \{W^1, b^1, W^2, b^2, \dots, W^L, b^L\}$$



$$f(x) \approx \sum_i [a_i \leq x \leq a_{i+1}] f(a_i)$$

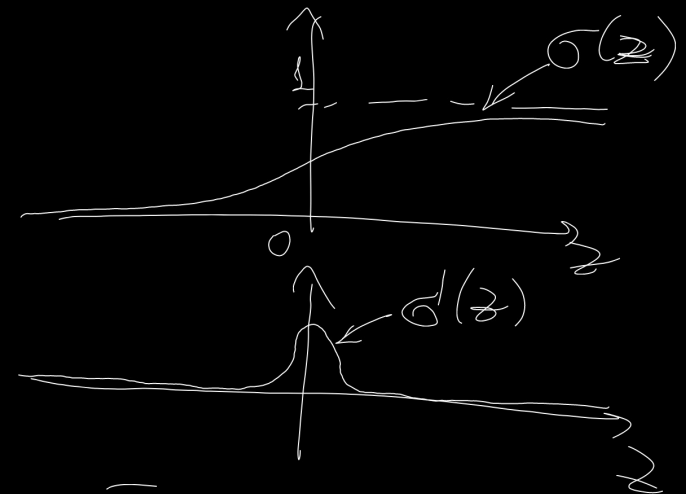
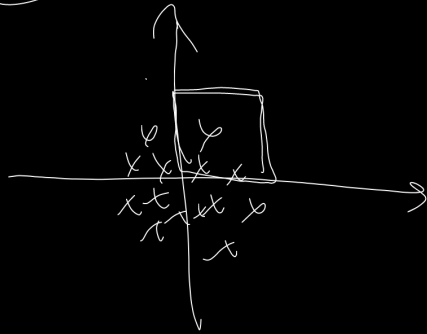
$$[x \geq a] \approx \delta(x - a) \quad x \gg 1$$



$$[a \leq x \leq b] = [x \geq a] + [x \leq b] \approx \frac{3}{2}$$

Activation functions

$$\textcircled{1} \sigma(z) = \frac{1}{1 + \exp(-z)}$$



$$\nabla_{a^k} \text{Loss} = \prod_{k=l+1}^L (W^k)^T \frac{\partial a_k}{\partial z_k} \nabla_{a^k} \text{Loss}$$

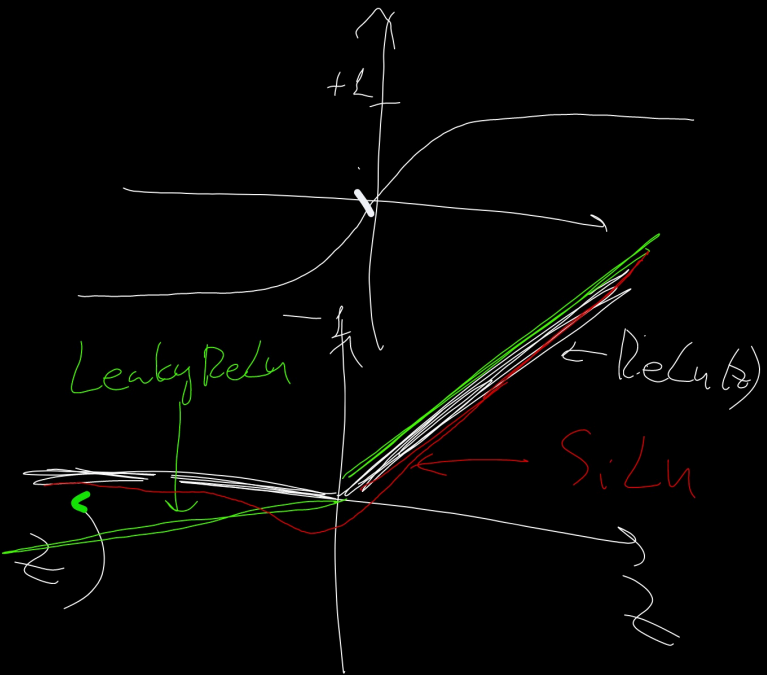
$$\|\nabla_{a^k} \text{Loss}\| \leq \prod_{k=l+1}^L \|W^k\| \cdot \|\text{diag}(\sigma'(z))\| \cdot \|\nabla_{a^k} \text{Loss}\|$$

$$\textcircled{2} \quad \tanh(z) = 2\sigma(2z) - 1$$

$$\textcircled{3} \quad \text{ReLU}(z) = \max(0, z)$$

$$\textcircled{4} \quad \text{Leaky ReLU}(z) = \max(\alpha z, z)$$

$$\textcircled{5} \quad \text{SiLU}(z) = z\sigma(z), \quad \alpha = 0.5, \quad \text{GELU}, \text{Swish}, \dots$$



Automatic Differentiation

$f(x)$

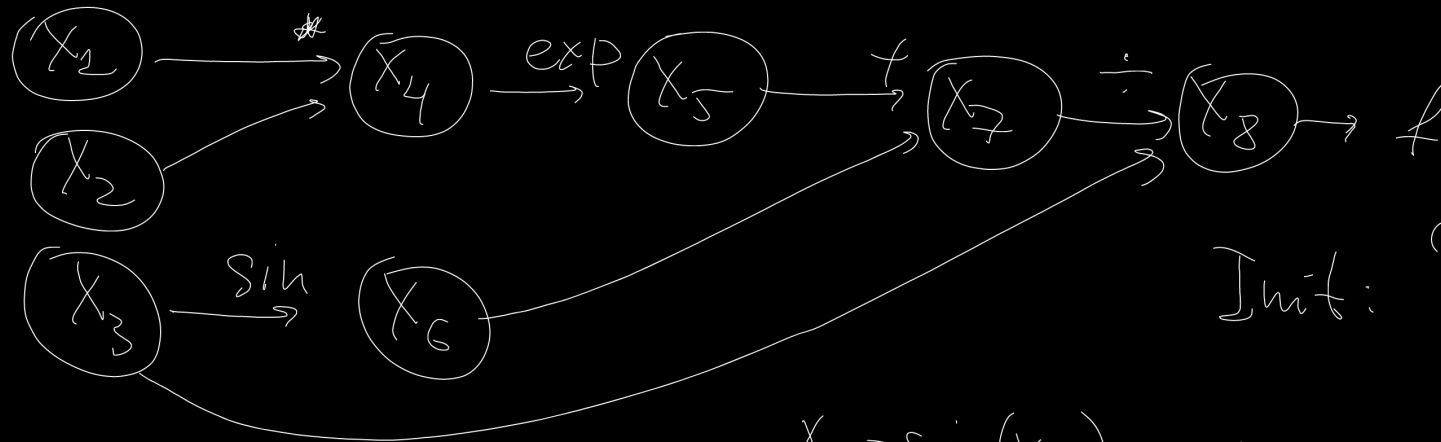
Goal: $\nabla_x f(x)$

$[0 \dots 0 \overset{i^{\text{th}} \text{ pos.}}{\underset{\uparrow}{1}} 0 \dots 0]$

$$\frac{\partial f}{\partial x_i} \approx \frac{f(x + \varepsilon e_i) - f(x - \varepsilon e_i)}{2\varepsilon}$$

$\varepsilon = \varepsilon_m^{1/3}$
 ε_m - machine precision

$$f(x_1, x_2, x_3) = \frac{\exp(x_1 x_2) + \sin(x_3)}{x_3}$$



Init: $\frac{\partial x_1}{\partial x_1} = 1, \frac{\partial x_2}{\partial x_1} = 0, \frac{\partial x_3}{\partial x_1} = 0$

$$x_6 = \sin(x_3)$$

$$\frac{\partial x_6}{\partial x_1} = \cos(x_3) \cdot \frac{\partial x_3}{\partial x_1}$$

$$x_4 = x_1 x_2$$

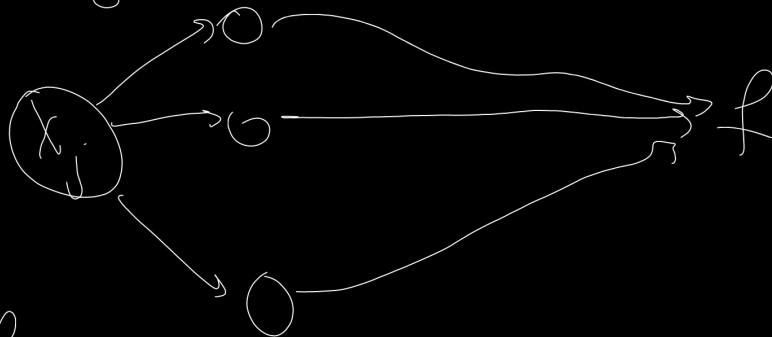
$$\frac{\partial x_4}{\partial x_1} = \frac{\partial x_1}{\partial x_1} x_2 + x_1 \frac{\partial x_2}{\partial x_1}$$

Forward mode

$$\frac{\partial x_j}{\partial x_1}, j=1, \dots, 8$$

Backward mode

$$\frac{\partial f}{\partial x_j}, \quad j=8, 7, \dots, 1$$



$$\frac{\partial f}{\partial x_j} = \sum_{i: (i,j) \in E} \frac{\partial f}{\partial x_i} \frac{\partial x_i}{\partial x_j}$$

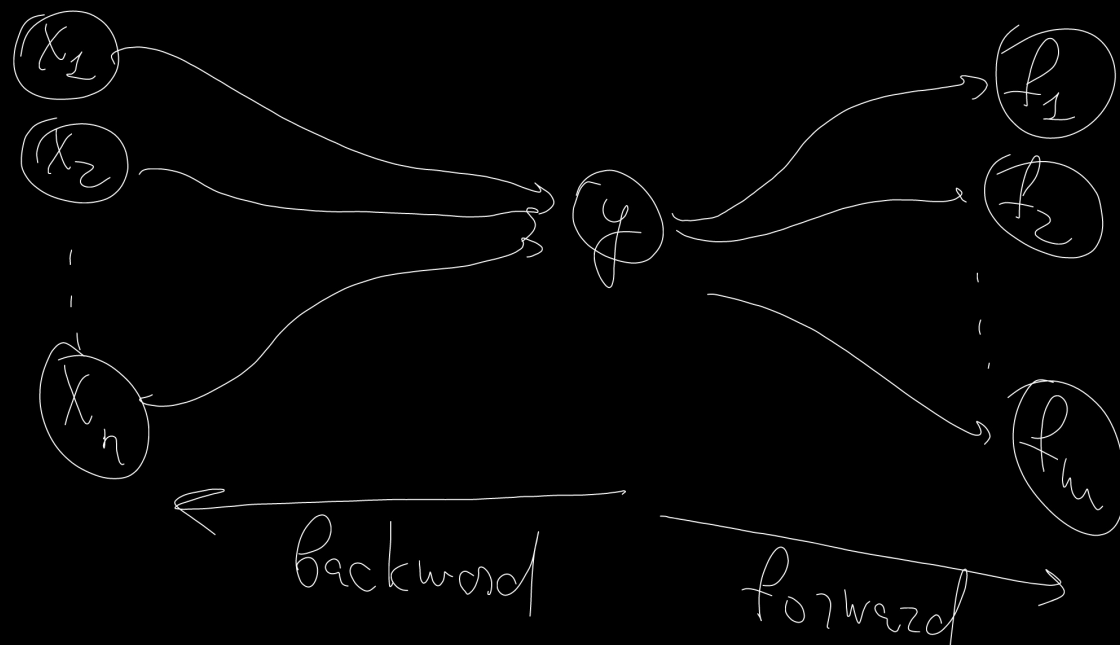
$$\text{Init: } \frac{\partial f}{\partial x_8} = 1$$

$$x_8 = \frac{x_7}{x_3}$$

$$\frac{\partial f}{\partial x_7} = \frac{\partial f}{\partial x_8} \cdot \frac{\partial x_8}{\partial x_7} = 1 \cdot \frac{1}{x_3}$$

Method	Memory	Computation
Forward mode	almost no	$\sim$ # input parameters
Backward mode	$\sim$ # internal nodes	$\sim$ 1 pass

Cross mode



$$\left\{ \frac{\partial f_i}{\partial x_j} \right\}_{i,j=1}^{m,n}$$

$$\frac{\partial f_i}{\partial x_j} = \frac{\partial f_i}{\partial y} \cdot \frac{\partial y}{\partial x_j}$$

$$f(X) = \text{tr}(A X^{-1} B) : \mathbb{R}^{n \times n} \rightarrow \mathbb{R}$$

$$\frac{\partial f}{\partial X} = \frac{\partial}{\partial X} \text{tr}(\dots) = \text{tr}\left(\frac{\partial}{\partial X} \underbrace{(A X^{-1} B)}_{n \times n}\right)$$

$$f: U \rightarrow V$$

Def. $f(x+h) - f(x) = \underbrace{df(x)[h]}_{\text{linear op. w.r.t. } h} + \mathcal{O}(\|h\|)$

4d tensor

$$f: \mathbb{R} \rightarrow \mathbb{R}$$

$$df = f'(x) dx$$

$$f: \mathbb{R}^n \rightarrow \mathbb{R}$$

$$df = \nabla f(x)^T dx$$

$$f: \mathbb{R}^{n \times m} \rightarrow \mathbb{R}$$

$$df = \text{tr}(\nabla_x f^T dx)$$

$$f(X) = \text{tr}(AX^{-1}B)$$

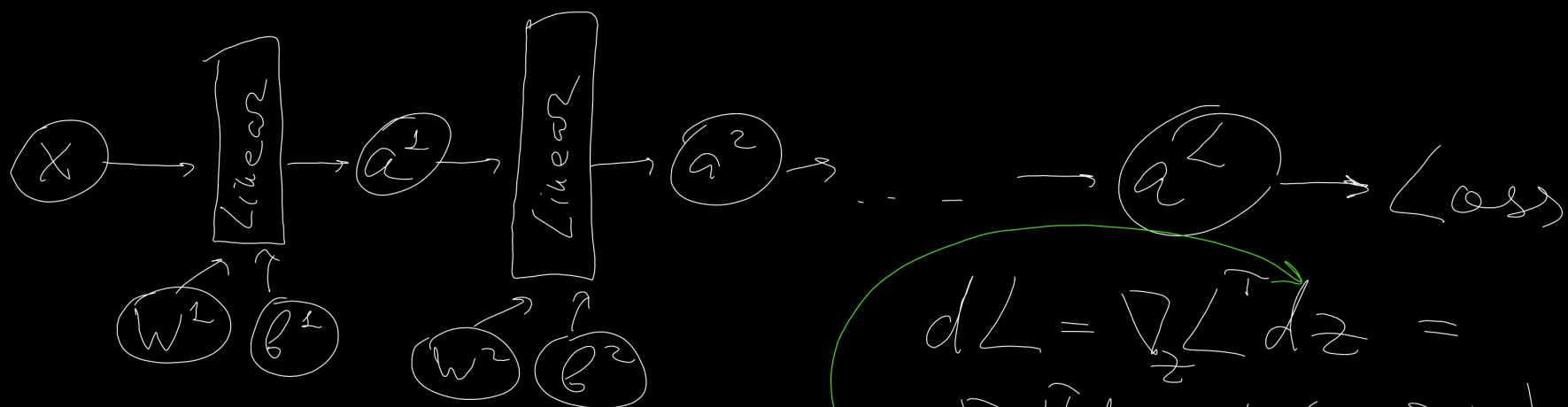
$$\begin{aligned} df &= \text{tr}(d(AX^{-1}B)) = \text{tr}(A \underbrace{dX^{-1}}_{-X^{-1}dX X^{-1}} B) = \\ &= -\text{tr}(A X^{-1} dX X^{-1} B) = -\text{tr}(X^{-1} B A X^{-1} dX) = \text{tr}(\nabla_X f^T dX) \\ &\Rightarrow \nabla_X f = -X^{-T} A^T B^T X^{-T} \end{aligned}$$

$$f(x), \quad \tilde{x} = g(x), \quad x = g^{-1}(\tilde{x}), \quad \tilde{f}(\tilde{x}) := f(g^{-1}(\tilde{x}))$$

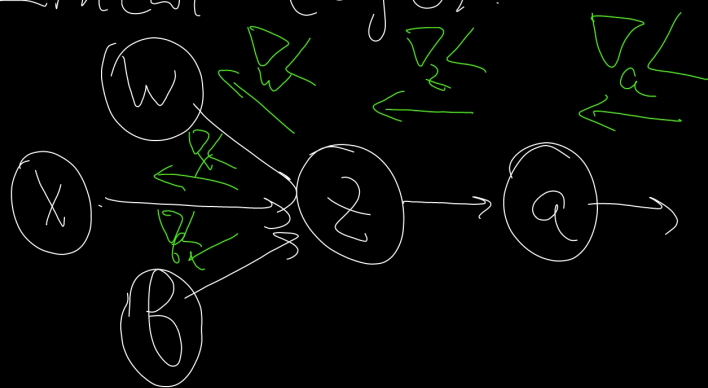
$\nearrow$
 one-to-one mapping

$$\nabla_x f \neq \nabla_{\tilde{x}} \tilde{f}$$

$$df = df(x)[dx] = d\tilde{f} = d\tilde{f}(\tilde{x})[d\tilde{x}]$$



Linear layer:



$$\begin{aligned}
 dL &= \nabla_z L^T dz = \\
 &= \nabla_b L^T db + \text{tr}(\nabla_w L^T dw) + \nabla_x L^T dx \\
 z &= wx + b \Rightarrow dz = dw \cdot x + w \cdot dx + db
 \end{aligned}$$

$$\begin{aligned}
 dL &= \nabla_z L^T (dw \cdot x + w dx + db) = \\
 &= \underbrace{\left(\nabla_z L^T dw \right)}_{\text{tr}(\dots dw)} + \underbrace{\nabla_z L^T w dx}_{\nabla_x L^T} + \underbrace{\nabla_z L^T db}_{\nabla_b L^T}
 \end{aligned}$$

$$\nabla_w L = \nabla_z L \cdot x^T$$

$$\nabla_x L = w^T \nabla_z L$$

$$\nabla_b L = \nabla_z L$$

$$a = g(z)$$

$$da = \frac{\partial g}{\partial z} dz$$

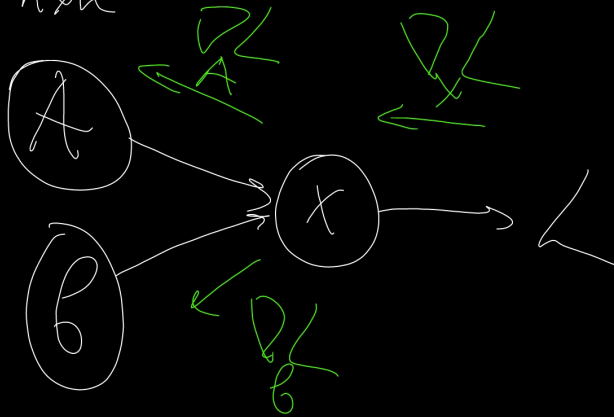
$$\text{diag}(g'(z))$$

$$\begin{aligned} dL &= \nabla_a L^T da = \\ &= \underbrace{\nabla_a L^T \frac{\partial g}{\partial z}}_{\nabla_z L^T} da \end{aligned}$$

$$\Rightarrow \nabla_z L = \begin{pmatrix} \frac{\partial g}{\partial z} \\ \frac{\partial g}{\partial a} \end{pmatrix}^T \nabla_a L$$

Example

$$Ax = b \Rightarrow x = A^{-1}b$$

$$h \times h$$


$$dL = \nabla_x L^T dx \quad (\textcircled{2})$$

$$Ax = b \quad | \quad d$$

$$dA \cdot x + A \cdot dx = d\ell$$

$$\Rightarrow dx = A^{-1}(dB - dA \cdot x)$$

$$\Rightarrow \mathcal{L}^T A^{-1} (d\ell - dA \cdot x) =$$